

Preliminary Results on Deep Sensor Fusion for Endpoint Prediction in Transhumeral Prostheses

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The majority of powered (robotic) upper-limb prostheses in clinical use are controlled via surface electromyography (sEMG) sensors worn on the residual limb. These prostheses are generally limited to sequential control, with the user controlling a single degree of freedom (DOF) at a time. This limitation arises primarily from difficulties in voluntarily controlling individual muscles, which are amplified by the level of the amputation, a particular challenge for transhumeral amputees.

Our objective is to develop a controller for an above-elbow prosthesis using alternative sensor modalities for predicting hand position, specifically gaze tracking and inertial measurement units (IMUs) in conjunction with sEMG. We believe such a predictor will enable simultaneous naturalistic control while reducing the mental load on users. To that end we have conducted healthy-subject pilot data acquisition combining sEMG, IMUs (head and arm mounted) and gaze-tracking, with optical motion capture (MoCap) for ground-truth kinematics. After pre-processing the data and transforming to a unified coordinate system, we calculate the elevation angles of both upper-arm segments according to Soechting and Flanders (1989). The first step of this work was to assess how effectively these sensor modalities can predict hand position, as such we assess the predictiveness of the elevation angles based on MoCap data rather than IMUs. Future implementations will use IMU data from the upper-arm to estimate elevation angles using a model robust to variations in IMU placement.

Our current deep-learning model is a hybrid CNN and GRU (Gated Recurrent Unit), which combines the fusion capabilities of a CNN with a GRU's ability to capture temporal phenomena. This architecture follows previous work where we quantified the temporal relationship between changes to a person's gaze trajectory and changes in the motion of their dominant hand. Our model is designed to predict the 3D coordinates of the palm and of optical markers mounted to either side of the wrist. After training, we tested our model on unseen data and achieved an average MAE of approximately 65mm across both hands. We believe this accuracy is quite good given the sparsity of the training data, as the average adult male hand width is 104mm wide according to Tilley *et al* (2001).

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References

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